

Human–Machine Interface in Orange Technology: A Care-Oriented Framework for Trust in International Medical Tourism

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Abstract— International medical tourism requires more than clinical excellence; it depends on trust fostered through technology-mediated communication. This study examines how Orange Technology, a care-oriented framework that integrates Human Machine Interface (HMI) with Human Computer Interaction (HCI) theory, promotes trust in international healthcare delivery. Focusing on Island Hospital in Penang, Malaysia, which serves predominantly Indonesian patients (representing 60% of international cases), we employed a qualitative methodology comprising systematic website observation, semi-structured interviews with 15 patients, 8 family members, and 3 Indonesian consulate officers, conducted between June – July 2024. Thematic analysis revealed five key trust-building mechanisms: (1) multilingual digital interfaces reducing language barriers, (2) transparent treatment and accommodation cost displays, (3) personalized pre-arrival online consultations, (4) culturally sensitive service desk interactions, and (5) continuous digital-physical care integration. These HMI applications operationalize Orange Technology principles by embedding empathy, cultural sensitivity, and humanistic values into both digital systems and human interactions. The study demonstrates that when properly designed, HMI extends beyond information delivery to create emotional connections, reduce patient anxiety, and foster sustainable trust in cross-border medical care. This research contributes to understanding how digital health technologies can advance patient well-being while maintaining human-centered care values.

Keywords—international medical tourism, orange technology, human machine interface, human computer interaction

I. INTRODUCTION

International medical tourism has become a rapidly expanding sector, driven by the increasing demand for affordable, high-quality healthcare beyond national borders. Countries in Southeast Asia, such as Malaysia, Thailand, and Singapore, have positioned themselves as leading destinations for international patients, offering competitive treatment packages, advanced medical technology, and comprehensive services that combine healthcare with hospitality [1]. In this context, trust emerges as the most crucial determinant of patients' decisions, shaping their

willingness to travel abroad for medical care. There are two trust categories, the medical profession and the specific medical practitioners [2]. Trust encompasses not only medical competence but also transparency, effective communication, and assurance of continuous support for patients and families throughout the care journey.

International medical tourism faces several barriers to obtain trust which can hinder patients' willingness to seek care abroad. Khan et al, categorized the risks associated with medical tourism into five main groups: (1) health-related risks at the destination; (2) risks stemming from extended air travel; (3) destination-specific risks; (4) medico-legal concerns; and (5) risks linked to the pre-operative phase and post-treatment recovery [3]. All of that creates the feeling of uncertainty for information transparency, language and cultural gaps, quality of care, and difficulty in maintaining communication with providers before traveling.

The digital era has reshaped this landscape, with technology playing an increasingly central role in fostering patient trust. The intersection of trust and technology is thought to be a current trend. This underscores the extent to which digital technology influences the formation and maintenance of individual trust, while simultaneously requiring that technology itself embody trust. These technologies serve as the first "frontline experience," mediating communication between patients and healthcare providers even before physical contact takes place [4].

Consequently, technology itself is no longer a neutral tool but must embody trustworthiness, empathy, and care to meet the expectations of international patients. Orange Technology represents a paradigm shift in health informatics, emphasizing the integration of humanistic values, emotional intelligence, and cultural sensitivity into technological systems [5]. Unlike traditional "Green Technology" (environmental sustainability) or "Blue Technology" (marine resources), Orange Technology prioritizes human well-being, happiness, and quality of life in the digital age. In healthcare contexts, Orange Technology manifests through systems that

not only deliver clinical information but also provide emotional support, cultural adaptation, and personalized care pathways [6].

The theoretical foundation of Orange Technology rests on three pillars:

1. Human-centeredness: Technology designed around user needs, emotions, and cultural contexts
2. Empathetic interaction: Systems that recognize and respond to emotional states and cultural preferences
3. Holistic well-being: Integration of physical health, emotional comfort, and social support

When applied to medical tourism, Orange Technology transforms HMI from mere information exchange tools into trust-building mechanisms that bridge cultural, linguistic, and emotional gaps between patients and foreign healthcare providers.

Against this backdrop, this study addresses the following research questions:

1. How does Orange Technology, through HMI applications, facilitate trust-building in international medical tourism?
2. What specific HMI features contribute to reducing uncertainty and enhancing patient confidence in cross-border healthcare?
3. How do Indonesian patients and their families perceive technology-mediated care at Island Hospital Penang?

The objectives of this study are to:

1. Examine the HMI applications employed by Island Hospital to serve Indonesian medical tourists
2. Analyze how these technologies embody Orange Technology principles of care-oriented design
3. Identify the trust-building mechanisms embedded in digital and human interactions
4. Provide recommendations for healthcare providers seeking to enhance cross-border patient trust through technology

Human Computer Interaction (HCI) theory provides the conceptual framework for understanding how users interact with digital systems, emphasizing usability, accessibility, and user experience [7]. Human Machine Interface (HMI) represents the practical implementation of HCI principles, the actual touchpoints where users engage with technology, including websites, mobile applications, kiosks, and chatbots [8].

In the context of Orange Technology, HMI serves as the vehicle through which care-oriented values are delivered. An HMI designed with Orange Technology principles does not simply present information; it anticipates patient anxiety, adapts to cultural communication styles, provides emotional reassurance, and creates pathways for personalized human connection when needed.

Island Hospital in Penang, Malaysia, provides a relevant case for examining this dynamic. Penang has emerged as one of Southeast Asia's leading medical tourism destinations, attracting patients from Indonesia, China, the Middle East,

and beyond. It stands out as the major contributor giving 60 percent of the country's medical tourist and revenue [9] and Island Hospital has been designated as one of four institutions under the Malaysian government's Flagship Medical Tourism Hospital Program (FMTH), aiming to deliver the Best Malaysia Healthcare Travel Experience by 2025 [10].

Island Hospital Penang was selected as the research site for several strategic reasons. First, it serves a predominantly Indonesian patient base, with Indonesian nationals comprising approximately 60% of its international patient volume [11]. Second, the hospital has invested significantly in culturally adapted HMI systems specifically designed for Southeast Asian patients. Third, its proximity to Indonesia and established referral networks make it an ideal case for studying technology-mediated trust in regional medical tourism. Finally, the hospital's integration of Orange Technology principles into both digital and physical care delivery provides a comprehensive example of care-oriented HMI implementation.

Building on this context, the study seeks to address the following main research question, how does the integration of orange Technology and Human-Machine Interface (HMI) in Island Hospital's digital services foster technology-mediated trust in international medical tourism? This question situates Island Hospital as a case study for understanding how care-oriented computing principles can enhance trust across borders. It also reflects the relevance of Human-Computer Interaction (HCI) theory, which emphasizes usability, transparency, and emotional resonance in the design of digital systems. By applying HCI theory, the study explores how the hospital's digital services—websites, chatbots, teleconsultations, and AI-driven communication tools—can be optimized not only for technical accuracy but also for relational care.

II. METHODOLOGY

The study employed a mixed qualitative approach comprising website observation and semi-structured interviews, conducted between June and October 2024. The website observation involved a systematic review of Island Hospital's official website, patient portal, mobile application, and social media channels. Attention was given to human-machine interaction features, multilingual accessibility, information transparency, cultural adaptation, and overall user experience. Screenshots, navigation flowcharts, and feature inventories were compiled to document the findings.

Semi-structured interviews were conducted with six Indonesian patients, eight accompanying family members, and three Indonesian Consulate officers in Penang. Using purposive sampling, participants were selected to reflect variation in treatment types, demographic profiles, and technology literacy. Interviews, lasting 30–60 minutes, were conducted in Bahasa Indonesia or English, either face-to-face or via video call. Discussions explored participants' information-seeking behavior, use of digital platforms, perceptions of trust, cultural and emotional experiences, and suggestions for improvement.

The interviews were conducted with the full consent of all participants, who were informed about the purpose and scope of the research prior to participation. Participants were assured that their responses would remain confidential and anonymous, and they were free to withdraw from the study at any time without consequence.

Data were analysed using thematic analysis guided by Braun and Clarke's six-phase framework. The process involved familiarization with the data through repeated reading of transcripts and notes, systematic coding of segments related to technology use, trust, and care experiences, and the development and refinement of themes. Themes were reviewed for coherence and clarity before being defined and illustrated with representative quotations.

To ensure trustworthiness, credibility was supported through member checking and data triangulation across sources, while transferability was strengthened by providing rich contextual descriptions. Dependability was maintained through an audit trail of analytical decisions, and confirmability was ensured through reflexive journaling to address potential researcher bias.

Data supporting this study is available from Zenodo <https://doi.org/10.5281/zenodo.18150762>. Access to the data is subjected to approval and a data sharing agreement due to the publisher policy.

III. RESULT

The analysis revealed five major themes illustrating how human-machine interfaces (HMI) operationalize Orange Technology principles to build trust in international medical tourism. Each theme is presented below with supporting evidence from website observations and interviews.

A. Multilingual Digital Interfaces as Linguistic Bridges

Island Hospital's website and patient portal provide comprehensive Bahasa Indonesia translations, extending beyond word-for-word conversion to culturally appropriate phrasing. Patients highlighted that the availability of content in their native language reduced anxiety and conveyed that the hospital understood their needs.

Supporting Evidence:

- Website observation: Homepage, treatment information pages, and patient portal all include Bahasa Indonesia options; key procedural and cost information is clearly structured.
- Patient quote: "When I first visited the website, I was relieved to see everything in Bahasa. It made me feel they understood Indonesian patients..." (P2, 58-year-old male, orthopedic patient)

This demonstrates that multilingual interfaces function as emotional as well as informational bridges, aligning with

HMI and HCI principles that reduce cognitive load and support trust formation.

B. Transparent Cost Information Reducing Financial Anxiety

Transparent presentation of treatment costs, accommodation packages, and doctor qualifications alleviated one of the most pressing concerns for medical tourists: financial uncertainty. Patients and family members reported feeling more confident to proceed with care when costs were clearly displayed.

Supporting Evidence:

- Website observation: Treatment packages include detailed pricing and options for ancillary services.
- Family and patient quote: "Seeing the costs upfront made planning easier. I didn't have to worry about hidden fees..." (F1 and F2, parent accompanying patient and P4, 36-year-old male, Otorhinolaryngology patient)

Transparency is not merely an informational feature but an enactment of Orange Technology principles, signaling respect and care for patient dignity.

C. Personalized Pre-Arrival Online Consultations

Pre-arrival consultations via WhatsApp, live chat, or video call allowed patients to clarify questions and plan their treatment remotely. This personalized interaction extended the hospital's interface into the patient's daily life and reduced anxiety before travel.

Supporting Evidence:

- Patient quote: "I could chat directly with my doctor online and get answers in my language. It made me feel safe even before leaving home..." (P1, 45-year-old female and P3, 60-year-old female, cancer post patient)

Such digital consultations exemplify how technology mediates emotional reassurance and builds relational trust.

D. Culturally Sensitive Service Desk Integration

The hospital provides dedicated Indonesian-language service desks and staff trained in regional dialects, ensuring continuity of care and minimizing cultural barriers. Patients reported feeling supported not only digitally but also physically upon arrival.

Supporting Evidence:

- Observation: Indonesian service desk signage and staff available on arrival.

- Patient quote: “The staff spoke my dialect and answered questions patiently. I immediately felt welcomed...” (F2, F3, F4, F5, range 45 - 55-year-old male and female, accompanying patient)

This integration of digital and human services reflects HMI design that considers emotional, linguistic, and cultural needs, reinforcing multi-dimensional trust.

E. Digital-Physical Care Continuity

Island Hospital’s coordination of accommodation, transportation, and online-offline service touchpoints created a seamless experience. Patients and families emphasized the importance of consistent messaging and support across digital and physical channels.

Supporting Evidence:

- Observation: Coordinated booking systems link hospital appointments, accommodations, and transport options.
- Family quote: “Everything from online booking to our room was smooth. It felt like the hospital had thought of everything...” (All patients’ family, accompanying patient)

By aligning technology with service processes, the hospital establishes trust as an ongoing, relational process rather than a one-time interaction.

F. Cross-Cutting Insight: Emotional Reassurance Through Design

Across all themes, digital interfaces were not only functional but emotionally expressive. Technology communicated attentiveness, care, and cultural awareness, showing that trust in medical tourism is both cognitive and affective.

IV. DISCUSSION

A. HMI as Emotional Infrastructure in Medical Tourism

The study shows that HMI applications in medical tourism serve as emotional infrastructure, bridging psychological and cultural distances. Multilingual websites, responsive communication tools, and culturally adapted interfaces functioned not just as information systems but as mechanisms for relational reassurance. For Indonesian patients, these features provided a sense of security and belonging in an otherwise unfamiliar healthcare environment.

B. Orange Technology in Practice: From Theory to Implementation

This research provides empirical evidence of Orange Technology principles in action. Transparent websites, personalized consultations, and family-inclusive digital services illustrate how technology can embody empathy, warmth, and relational care. Island Hospital operationalizes

these principles by designing interfaces that attend simultaneously to informational, emotional, and cultural needs.

C. Trust as a Multi-Dimensional Construct

Trust emerges as a multi-layered construct shaped by the integration of digital and human touchpoints. Transparent cost information, multilingual access, personalized communication, culturally sensitive desks, and coordinated offline support collectively foster confidence. This finding underscores that trust in medical tourism is sustained not only by the availability of information but also by the relational and emotional quality of interactions.

D. Implications for Healthcare Providers

Healthcare providers aiming to attract international patients should:

1. Develop multilingual and culturally contextualized digital platforms.
2. Offer multiple responsive communication channels to support emotional reassurance.
3. Ensure continuity between digital services and offline care, including culturally competent staff and logistical support.

E. Theoretical Contributions

This study extends HCI and healthcare technology theory by showing that digital platforms can serve as care-oriented instruments. Cultural adaptation, emotional design, and institutional alignment collectively mediate patient trust, demonstrating that technology in healthcare is both functional and relational.

F. Limitations

Limitations include the single-case study design, focus on Indonesian patients, potential positivity bias, cross-sectional data, and occasional language barriers during interviews. These factors constrain generalizability and temporal analysis of trust development.

G. Future Research Directions

Future studies should examine multiple hospitals across countries, employ quantitative measures of trust, conduct longitudinal research on trust evolution, investigate negative or failed cases, and explore the role of social media and online patient communities in shaping trust.

The discussion based on the results of this study are organized into four interrelated themes: (1) website transparency, (2) communication tools, (3) patient and family experiences, and (4) institutional support. Together, these dimensions demonstrate how digital tools—when designed using Human–Machine Interface (HMI) and Human–Computer Interaction (HCI) principles and guided by Orange Technology—mediate trust in international medical tourism. The results show that technology in healthcare functions not only as a medium of efficiency but also as an infrastructure of warmth, relationality, and emotional assurance, which are

critical when patients and families face heightened vulnerability.

1. Website Transparency

Island Hospital's multilingual website was consistently identified by participants as a central trust-building mechanism. The platform provides transparent information on physician qualifications, treatment procedures, accommodation options, and estimated costs, thereby addressing one of the main anxieties of medical travelers—hidden expenses and uncertain service quality. The inclusion of Bahasa Indonesia content was particularly meaningful to Indonesian patients, who constitute nearly 90% of the hospital's international clientele.

From an HMI standpoint, the hospital's website serves as the primary interface mediating initial contact between potential patients and the institution. Its usability, clarity, and linguistic accessibility are not merely technical attributes but essential determinants of credibility and trustworthiness. According to HCI theory, interfaces that minimize cognitive load and uncertainty tend to enhance user trust and long-term engagement. By providing comprehensive and transparent information, Island Hospital effectively reduces informational asymmetry, a common source of anxiety in cross-border healthcare.

Within the framework of Orange Technology, transparency itself represents a form of humanistic care. Explicit cost disclosure and service visibility communicate respect for patient dignity and acknowledgment of their vulnerabilities. The hospital's website therefore functions not merely as an informational portal but as a technological embodiment of care, reinforcing patient agency and emotional security prior to treatment.

2. Communication Tools

The second major finding concerns the hospital's extensive range of communication platforms, including WhatsApp, email, live chat, online consultations, chatbots, and direct messaging with doctors. Patients emphasized that these tools' responsiveness and flexibility provided reassurance and facilitated treatment decision-making.

From an HMI perspective, these platforms extend the hospital's interface into the patient's daily life, forming a seamless digital bridge across geographical distance. Consistent with HCI theory, adaptive and multimodal communication systems increase personalization and foster trust by accommodating users' preferred modes of interaction. Importantly, participants did not perceive these technologies as impersonal or transactional; instead, they described them as channels of personalized care due to the immediacy of responses and the empathetic tone of communication.

This pattern reflects the Orange Technology principle of emotional warmth. By adopting culturally familiar

communication tools such as WhatsApp, Island Hospital demonstrated attentiveness to emotional comfort as well as operational efficiency. Patients reported that being able to text doctors directly or consult via accessible mobile apps created a sense of proximity and relational trust. Technology, therefore, becomes more than infrastructure—it functions as a symbol of relational care that communicates attentiveness and commitment to patient well-being.

3. Patient and Family Experiences

Interviews with patients and their families indicated that trust was cultivated not only through access to accurate information but also through the hospital's capacity to deliver culturally sensitive and family-inclusive care. Respondents highlighted the importance of native-language communication, the availability of an Indonesian service desk, and assistance with accommodation and transport—all of which reduced logistical burdens and emotional stress during their stay in Penang.

From an HMI and HCI perspective, these findings underscore that user-centered design extends beyond the digital interface into the broader patient journey. By integrating Bahasa Indonesia services and logistical support, Island Hospital generated a sense of belonging and emotional security within a foreign environment. The hospital thus expanded the locus of trust from the individual patient to the entire family unit.

Aligned with Orange Technology, the institution's digital ecosystem integrates functional efficiency with empathy and social responsibility. Patients valued staff and doctors who communicated in their preferred languages and displayed interpersonal warmth, transforming potentially intimidating medical encounters into trustworthy relationships. Furthermore, linguistic adaptability extended to digital communication, with staff trained in regional dialects such as Indonesian Mandarin. These practices embody a Care Service-Oriented approach, where both digital and human touchpoints are intentionally designed to nurture emotional closeness and cultural alignment.

4. Institutional Support

Institutional support from both Island Hospital and the Penang government reinforced the trust cultivated through digital interfaces and interpersonal communication. Dedicated Indonesian-language desks, multilingual staff, partnerships with Indonesian travel agents, and integrated accommodation and transport services created a holistic trust environment. Crucially, these offline services were consistent with the hospital's online information, ensuring alignment between expectation and experience.

This reflects the concept of Care Service-Oriented Computing, in which organizational systems are structured to provide seamless and reliable patient experiences. From an HMI and HCI perspective, such integration ensures

continuity between digital and physical interactions, enabling sustainable, technology-mediated trust.

Institutional strategies also embody the vision of Orange Technology, which emphasizes responsibility, warmth, and human-centered design. By institutionalizing cultural sensitivity and embedding emotional care into its operational framework, Island Hospital moves beyond transactional service delivery toward systemic, trust-based healthcare engagement.

Overall, the findings highlight that technology in medical tourism functions as both emotional infrastructure and institutional connector. Trust emerges not from digital efficiency alone but from the seamless interplay between cultural understanding, emotional resonance, and consistent service delivery.

For healthcare providers, these insights underscore the importance of designing multilingual, emotionally aware, and contextually adaptive systems that merge digital and human care. In practice, the success of HMI in international healthcare depends not only on usability or interface quality but also on its capacity to embody care, empathy, and cross-cultural understanding.

ACKNOWLEDGMENT

This work was supported by Bina Nusantara University as part of the BINUS International Research-Basic program, project entitled "Medical Tourism from the Perspective of Communication: A Case Study of Indonesian Patients in Penang Malaysia" (Contract No. 069B/VRRTT/III/2024, dated March 18, 2024).

The authors gratefully acknowledge Island Hospital Penang for facilitating access to participants and the Indonesian Consulate in Penang for their cooperation. We also extend our sincere thanks to all patients and family members who generously shared their experiences. Special appreciation to [research assistants' names] for their assistance with data collection and transcription.

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